

25. Ultra-Low-Power Timer (ULPT)

25.1 Overview

This ultra-low power timer (ULPT) is a 32-bit timer that can output pulses or count external events.

This 32-bit timer consists of reload registers and a down-counter. The reload registers and the down-counter are allocated to the same address and can be accessed through the ULPTCNT register.

Table 25.1 lists the ULPT specifications.

Table 25.1 ULPT specifications

Item		Description
Operating mode	Timer mode	Count the internal clock
	Event counter mode	Count external events
Number of channels		32 bits × 2 channels (ULPTn (n = 0, 1))
Count source (operating clock)*1		The following signals can be selected: - Internal clock: ULPTLCLK/d or ULPTSCLK/d (d = 1, 2, 4, 8, 16, 32, 64, 128) - External event: ULPTVIN pin.
Count operation		The following functions can be selected: - Selection between continuous mode and one-shot mode - Selection from count enable mode, count start mode, and count restart mode.
Interrupt/event link function (output)		- Underflow signal - When the counter underflows - Compare match A signal - When the ULPTCNT value matches the ULPTCMA value - The compare match A function is enabled. - Compare match B signal - When the ULPTCNT value matches the ULPTCMB value - The compare match B function is enabled.
Selectable functions		- Pulse output function - Inverts the output each time the counter underflows - Compare match function - Either or both compare match A register and compare match B register are selectable - Return from Software Standby mode can be performed with ULPTn_ULPTI, ULPTn_ULPTCMAI, or ULPTn_ULPTCMBI (n = 0, 1).^{*2} - Return from Deep Software Standby mode 1 can be performed with ULPTn_ULPTI (n = 0, 1).^{*2}
Module-stop function		Module-stop state can be set for each channel to reduce power consumption
TrustZone Filter		Security and privilege attribution can be set for each channel

Note 1. Make sure that the frequency of the peripheral module clock (PCLKB) is at least twice the frequency of the count source clock.

Note 2. For details, see [section 11, Low Power Mode](#).

25.2 Configuration

Figure 25.1 shows the ULPT block diagram. Table 25.2 lists the I/O pins of the ULPT.

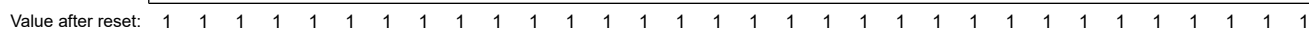


Pin name	I/O	Function
ULPTEEn*2	Input	External count control input*1
ULPTEVIn*2	Input	External event input
ULPTOn*2	Output	Pulse output
ULPTOAn*2	Output	Output compare match A output
ULPTOBn*2	Output	Output compare match B output

Note 2. Only pins marked "-DS" can be used in Deep Software Standby mode 1.

25.3.1 ULPTCNT : ULPT Counter Register

0



Bit	Symbol	Function	R/W
31:0	ULPTCNT	32-bit counter and reload register ^{*1*2} Setting range : 0x00000000 to 0xFFFFFFFF	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. When 1 is written to the ULPTCR.TSTOP bit, the counter is forcibly stopped and set to 0xFFFFFFFF. The reload register is set to 0xFFFFFFFF.

Note 2. When 0x00000000 is set in the ULPTCNT register, a request signal to the ICU, DTC, and ELC occurs only once immediately after the start of count operation (when the ULPTCR.TSTART bit is set to 1). However, the pulse output at the ULPTOn pin remains toggled.

The pulse output from the ULPTOn pin can be enabled or disabled by the ULPTIOC.TOE bit.

When the ULPTCNT register is set to a value equal to or greater than 0x00000001, a request signal is generated each time the counter underflows.

The ULPTCNT is a 32-bit register. The register value is written to the reload register and is read from the counter.

The states of the counter and reload register change according to the TSTART/TCSTF bit in the ULPTCR register and the TCMEA/TCMEB bit in the ULPTCMSR register. For details, see [section 25.4.2. Rewriting the Counter and Reload Register](#).

The ULPTCNT register is only available for 32-bit read/write access.

25.3.2 ULPTCMA : ULPT Compare Match A Register

Base address: ULPTn = 0x4022_0000 + 0x0100 × n (n = 0, 1)
ULPTn_NS = 0x5022_0000 + 0x0100 × n (n = 0, 1)

Offset address: 0x04

Bit position: 31

0

Bit field:

ULPTCMA

Value after reset: 1

Bit	Symbol	Function	R/W
31:0	ULPTCMA	32-bit Compare Match A Data ^{*1*2} Setting range : 0x00000000 to 0xFFFFFFFF	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. When 1 is written to the ULPTCR.TSTOP bit, 0xFFFFFFFF is set in the ULPTCMA register.

Note 2. Set the ULPTCMA register to 0xFFFFFFFF when compare match A is not used (ULPTCMSR.TCMEA bit = 0).

The ULPTCMA register is a readable/writable register to set a value for compare match with the counter.

The states of compare circuit A and the reload register change according to the TSTART and TCSTF bits in the ULPTCR register. For details, see [section 25.4.3. Rewriting the Compare Circuits and Reload Registers for Compare Match A/B](#).

The ULPTCMA register is only available for 32-bit read/write access.

25.3.3 ULPTCMB : ULPT Compare Match B Register

Base address: ULPTn = 0x4022_0000 + 0x0100 × n (n = 0, 1)
ULPTn_NS = 0x5022_0000 + 0x0100 × n (n = 0, 1)

Offset address: 0x08

Bit position: 31

0

Bit field:

ULPTCMB

Value after reset: 1

Bit	Symbol	Function	R/W
31:0	ULPTCMB	32-bit Compare Match B Data ^{*1*2} Setting range : 0x00000000 to 0xFFFFFFFF	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. When 1 is written to the ULPTCR.TSTOP bit, 0xFFFFFFFF is set in the ULPTCMB register.

Note 2. Set the ULPTCMB register to 0xFFFFFFFF when compare match B is not used (ULPTCMSR.TCMEB bit = 0).

The ULPTCMB register is a readable/writable register to set a value for compare match with the counter.

The states of compare circuit B and the reload register change according to the TSTART and TCSTF bits in the ULPTCR register. For details, see [section 25.4.3. Rewriting the Compare Circuits and Reload Registers for Compare Match A/B](#).

The ULPTCMB register is only available for 32-bit read/write access.

25.3.4 ULPTCR : ULPT Control Register

Base address: $ULPTn = 0x4022_0000 + 0x0100 \times n$ ($n = 0, 1$)
 $ULPTn_NS = 0x5022_0000 + 0x0100 \times n$ ($n = 0, 1$)

Offset address: 0x0C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TCMB F	TCMA F	TUNDF F	—	—	TSTOP P	TCSTF F	TSTART RT
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TSTART	Counter Start 0: Stop the counter 1: Start the counter	R/W
1	TCSTF	Counter Status Flag* ² 0: Counter stopped 1: Counter running	R
2	TSTOP	Counter Forcible Stop* ¹ 0: Writing is invalid 1: Stop the counter forcibly	W
4:3	—	These bits are read as 0. The write value should be 0.	R/W
5	TUNDF	Underflow Flag 0: No underflow occurred (counter $\neq$ 0x00000000) 1: Underflow occurred (counter = 0x00000000)	R/W* ³
6	TCMAF	Compare Match A Flag 0: Not matched (counter $\neq$ ULPTCMA[31:0]) 1: Matched (counter = ULPTCMA[31:0])	R/W* ³
7	TCMBF	Compare Match B Flag 0: Not matched (counter $\neq$ ULPTCMB[31:0]) 1: Matched (counter = ULPTCMB[31:0])	R/W* ³

Note: S-TYPE-3, P-TYPE-3

Note 1. When 1 (stop the counter forcibly) is written to the TSTOP bit, the TSTOP, TSTART, and TCSTF bits are initialized. The pulse output, compare match A output and compare match B output are also initialized. This bit is read as 0.

Note 2. For precautions on using the TCSTF bit, see [section 25.5.1. Start and Stop Control of the Counter](#).

Note 3. Only 0 can be written to clear the flag.

The ULPTCR register controls the start and stop of the counter and indicates the status of the ULPT.

The ULPTCR register is only available for 8-bit read/write access.

Set 1 to the TUNDF, TCMAF, and TCMBF bits when changing the TSTART or TSTOP bit to prevent clearing TUNDF, TCMAF, or TCMBF by mistake. At this time, setting 1 to TUNDF, TCMAF, and TCMBF bits is ignored.

TSTART bit (Counter Start)

When and how the counter is started depends on the setting of the ULPTMR3.TEECTL[1:0] bits.

The counter is stopped in the same way in any mode.

For details, see [section 25.4.1. Count Operation](#).

[Starting the counter]

In count enable mode (ULPTMR3.TEECTL[1:0] bits = 00)

- When 1 is written to the TSTART bit.

In count start mode or count restart mode (ULPTMR3.TEECTL[1:0] bits = 10 or 11)

- When 1 is written to the TSTART bit, and then an edge at the ULPTEEn pin is input.

[Stopping the counter]

- When 0 is written to the TSTART bit
- When 1 is written to the TSTOP bit.

TCSTF bit (Counter Status Flag)

The operation of this bit depends on the settings of the TCNTCTL and TEECTL[1:0] bits of the ULPTMR3 register.

For details, see [section 25.4.1. Count Operation](#).

[Setting conditions]

In timer mode (ULPTMR1.TMOD1 = 0) and count enable mode (ULPTMR3.TEECTL[1:0] bits = 00)

- When 1 is written to the TSTART bit (with the TCSTF bit set to 1 in synchronization with the count source).

In event counter mode (ULPTMR1.TMOD1 = 1) and count enable mode (ULPTMR3.TEECTL[1:0] bits = 00)

- When 1 is written to the TSTART bit (with the TCSTF bit set to 1 in synchronization with the count source).

In count start mode or count restart mode (ULPTMR3.TEECTL[1:0] bits = 10 or 11)

- When 1 is written to the TSTART bit, and then an edge at the ULPTEEn pin is input (with the TCSTF bit set to 1 in synchronization with the count source).

[Clearing conditions]

- When 0 is written to the TSTART bit (with the TCSTF bit set to 0 in synchronization with the count source)
- When 1 is written to the TSTOP bit
- When the counter underflows in one-shot mode (ULPTMR3.TCNTCTL = 1).
Only under this condition, the TSTART bit does not change from 1.

TSTOP bit (Counter Forcible Stop)

When 1 is written to this bit, the counter is forcibly stopped. This bit is read as 0.

TUNDF bit (Underflow Flag)

[Setting condition]

- When the counter underflows (counter = 0x00000000).

[Clearing condition]

- When 0 is written to this bit.

TCMAF bit (Compare Match A Flag)

[Setting condition]

- When the ULPTCNT register value matches the ULPTCMA register value (counter = ULPTCMA[31:0]).

[Clearing condition]

- When 0 is written to this bit.

TCMBF bit (Compare Match B Flag)

[Setting condition]

- When the ULPTCNT register value matches the ULPTCMB register value (counter = ULPTCMB[31:0]).

[Clearing condition]

- When 0 is written to this bit.

25.3.5 ULPTMR1 : ULPT Mode Register 1

Base address: $ULPTn = 0x4022_0000 + 0x0100 \times n$ ($n = 0, 1$)
 $ULPTn_NS = 0x5022_0000 + 0x0100 \times n$ ($n = 0, 1$)

Offset address: 0x0D

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	TCK1	—	TEDGPL	—	TMOD1	—

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	TMOD1	Operating Mode* ¹ 0: Timer mode 1: Event counter mode	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	TEDGPL	ULPTEVIn Edge Polarity 0: Either edge (rising) 1: Both edges This bit is valid only in event counter mode (TMOD1 = 1).	R/W
4	—	This bit is read as 0. The write value should be 0.	R/W
5	TCK1	Count Source* ² 0: Divided clock specified by the ULPTMR2.CKS[2:0] bits (ULPTLCLK) 1: Divided clock specified by the ULPTMR2.CKS[2:0] bits (ULPTSCLK) This bit is valid only in timer mode (TMOD1 = 0).	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Do not rewrite this register while the counter is running. Rewrite this register only when the counter is stopped (both the TSTART and TCSTF bits of the ULPTCR register are set to 0).

Note: A write access to this register initializes the output from pins ULPTOn, ULPTOAn, and ULPTOBn. For details on the output level at initialization, see [section 25.3.7. ULPTMR3 : ULPT Mode Register 3](#) and [section 25.3.8. ULPTIOC : ULPT I/O Control Register](#).

Note 1. Timer mode: The internal clock (ULPTLCLK or ULPTSCLK) is used as the count source. The counter is decremented by 1 on each rising edge of the count source.

Event counter mode: The external event input (ULPTEVIn pin) is used as the count source. The counter is decremented on each edge of the count source. The edge polarity can be selected by the TEDGPL bit and the ULPTMR3.TEVPOL bit.

For details, see [section 25.4.1. Count Operation](#).

Note 2. The TCK1 bit can be changed only when the ULPTMR2.CKS[2:0] bits are 000. Otherwise, do not change the bit. Set the ULPTMR2.CKS[2:0] bits to 000, and then change the TCK1 bit after one cycle of the count source.

The ULPTMR1 register sets the operating mode of the ULPT.

The ULPTMR1 register is only available for 8-bit read/write access.

25.3.6 ULPTMR2 : ULPT Mode Register 2

Base address: $ULPTn = 0x4022_0000 + 0x0100 \times n$ ($n = 0, 1$)
 $ULPTn_NS = 0x5022_0000 + 0x0100 \times n$ ($n = 0, 1$)

Offset address: 0x0E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	LPM	—	—	—	—	CKS[2:0]		

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
2:0	CKS[2:0]	ULPTCLK/ULPTSCLK Count Source Clock Division Ratio ^{*1*2} 0 0 0: 1/1 0 0 1: 1/2 0 1 0: 1/4 0 1 1: 1/8 1 0 0: 1/16 1 0 1: 1/32 1 1 0: 1/64 1 1 1: 1/128 This bit is valid only in timer mode (ULPTMR1.TMOD1 = 0).	R/W
6:3	—	These bits are read as 0. The write value should be 0.	R/W
7	LPM	Low Power Mode 0: Normal mode 1: Low power mode	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Do not rewrite this register while the counter is running. Rewrite this register only when the counter is stopped (both the TSTART and TCSTF bits of the ULPTCR register are set to 0).

Note 1. The CKS[2:0] bits can be changed only in timer mode (ULPTMR1.TMOD1 = 0).

Set the CKS[2:0] bits to 000 in event counter mode (ULPTMR1.TMOD1 = 1).

Note 2. The ULPTMR1.TCK1 bit can be changed only when the CKS[2:0] bits are 000. Set the CKS[2:0] bits to 000, and then change the ULPTMR1.TCK1 bit after one cycle of the count source.

The ULPTMR2 register sets the operating mode of the ULPT.

The ULPTMR2 register is only available for 8-bit read/write access.

LPM bit (Low Power Mode)

This bit implements low power consumption by restricting register access. Set this bit to 1 for low-power mode operation.

When this bit is 1, access to the following registers is prohibited:

- ULPTCNT, ULPTCMA, ULPTCMB, and ULPTCR.

After this bit is switched from 1 to 0, the first access to an access-prohibited register is restricted as follows:

- ULPTCNT: The register must be read twice. Only the second reading of data is valid.
- ULPTCNT, ULPTCMA, ULPTCMB, ULPTCR: In write access, leave two cycles of the count source.

25.3.7 ULPTMR3 : ULPT Mode Register 3

Base address: ULPTn = 0x4022_0000 + 0x0100 × n (n = 0, 1)
ULPTn_NS = 0x5022_0000 + 0x0100 × n (n = 0, 1)

Offset address: 0x0F

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TEEPOL[1:0]	TEECTL[1:0]	—	TOPO L	TEVP OL	TCNT CTL		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TCNTCTL	Count Function Select ^{*1*2} 0: Continuous mode 1: One-shot mode	R/W
1	TEVPOL	ULPTEVIn Polarity Switch 0: External event input (ULPTEVIn pin) 1: External event input (ULPTEVIn pin) in reverse This bit is valid only in event counter mode (ULPTMR1.TMOD1 = 1).	R/W
2	TOPOL	ULPTOn Polarity Select ^{*3} 0: Start the ULPTOn output when in low level 1: Start the ULPTOn output when in high level	R/W

Bit	Symbol	Function	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
5:4	TEECTL[1:0]	ULPTEEn Function Select*1 0 0: Count enable mode 1 0: Count start mode 1 1: Count restart mode Other settings are prohibited. The count enable mode is valid only in event counter mode (ULPTMR1.TMOD1 = 1).	R/W
7:6	TEEPOL[1:0]	ULPTEEn Edge Polarity Select 0 0: Rising edge 0 1: Falling edge 1 0: Both edges Other settings are prohibited. This bit is valid only in count start mode or count restart mode. (TEECTL[1:0] = 10 or 11)	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Do not rewrite this register while the counter is running. Rewrite this register only when the counter is stopped (both the TSTART and TCSTF bits of the ULPTCR register are set to 0).

Note 1. For details, see [section 25.4.1. Count Operation](#).

Note 2. Continuous mode: After starting counting, the counter continues to count down (reload value to underflow) repeatedly until the counter stops.

One-shot mode: After starting counting, the counter counts down once (reload value to underflow).

Note 3. When data is set to the ULPTMR1 register or 1 is set to the ULPTCR.TSTOP bit, the ULPTOn output is initialized.

The ULPTMR3 register sets the operating mode of the ULPT.

The ULPTMR3 register is only available for 8-bit read/write access.

TEECTL[1:0] bits (ULPTEEn Function Select)

The ULPTEEn pin functions differently depending on the mode. For details, see [section 25.4.1. Count Operation](#).

In count enable mode, external events are counted while the ULPTEEn pin is valid. The duration can be selected by the ULPTIOC.TIOGT0 bit, and the polarity can be selected by the ULPTISR.RCCPSEL2 bit.

In count start mode, the counter starts counting at an edge trigger at the ULPTEEn pin. The operation of the ULPTEEn pin differs depending on the TSTART and TCSTF bits of the ULPTCR register.

In count restart mode, the counter starts counting at the first edge trigger at the ULPTEEn pin and resets the count at the second or following edge trigger. The operation of the ULPTEEn pin differs depending on the TSTART and TCSTF bits of the ULPTCR register.

[Table 25.3](#) shows the operation of the ULPTEEn pin in count start or count restart mode.

Table 25.3 Operation of the ULPTEEn pin in count start mode or count restart mode

TEECTL[1:0]	ULPTCR.TSTART	ULPTCR.TCSTF	Function of the ULPTEEn pin and counter operation
10 (count start mode)	0 (stop the counter)	—	The edge input at the ULPTEEn pin is invalid. The counter remains stopped.
	1 (start the counter)	0 (counter stopped)	The counter starts counting from the reload value on the edge input at the ULPTEEn pin.
	1 (start the counter)	1 (counter running)	The edge input at the ULPTEEn pin is invalid. The counter continues to count.
11 (count restart mode)	0 (stop the counter)	—	The edge input at the ULPTEEn pin is invalid. The counter remains stopped.
	1 (start the counter)	0 (counter stopped)	The counter starts counting from the reload value on the edge input at the ULPTEEn pin.
	1 (start the counter)	1 (counter running)	The counter is reset on the edge input at the ULPTEEn pin, and it starts counting again from the reload value.

Note: The edge polarity of the ULPTEEn pin can be selected by the TEEPOL[1:0] bits.

25.3.8 ULPTIOC : ULPT I/O Control Register

Base address: $ULPTn = 0x4022_0000 + 0x0100 \times n$ ($n = 0, 1$)
 $ULPTn_NS = 0x5022_0000 + 0x0100 \times n$ ($n = 0, 1$)

Offset address: 0x10

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	TIOGT 0	TIPF[1:0]	—	TOE	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W
2	TOE	ULPTOn Output Enable 0: Disable the ULPTOn output 1: Enable the ULPTOn output	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
5:4	TIPF[1:0]	ULPTEVIn Input Filter*2 0 0: No filter 0 1: Filter sampling at PCLKB 1 0: Filter sampling at PCLKB/8 1 1: Filter sampling at PCLKB/32 This bit is valid only in event counter mode (ULPTMR1.TMOD1 = 1).	R/W
6	TIOGT0	ULPTEVIn Count Control*1 0: Always count external events 1: Count external events while the ULPTEVIn pin is valid This bit is valid only in event counter mode (ULPTMR1.TMOD1 = 1) and count enable mode (ULPTMR3.TEECTL[1:0] = 00).	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Do not rewrite this register while the counter is running. Rewrite this register only when the counter is stopped (both the TSTART and TCSTF bits of the ULPTCR register are set to 0).

Note 1. The polarity for counting external events can be selected by the ULPTISR.RCCPSEL2 bit.

Note 2. Set the TIPF[1:0] bits to 00 before entering Software Standby mode or Deep Software Standby mode 1.

The ULPTIOC register controls the input/output of the ULPT.

The ULPTIOC register is only available for 8-bit read/write access.

TIPF[1:0] bits (ULPTEVIn Input Filter)

These bits are used to specify the sampling frequency of the ULPTEVIn input filter. Input to the ULPTEVIn pin is sampled. When a value matches three consecutive times, the value is regarded as the input value.

25.3.9 ULPTISR : ULPT Event Pin Select Register

Base address: $ULPTn = 0x4022_0000 + 0x0100 \times n$ ($n = 0, 1$)
 $ULPTn_NS = 0x5022_0000 + 0x0100 \times n$ ($n = 0, 1$)

Offset address: 0x11

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	RCCP SEL2	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
2	RCCPSEL2	ULPTEEn Polarity Select 0: Count external events when low 1: Count external events when high This bit is valid only in event counter mode (ULPTMR1.TMOD1 = 1), count enable mode (ULPTMR3.TEECTL[1:0] = 00), and when ULPTIOC.TIOGT0 = 1.	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Do not rewrite this register while the counter is running. Rewrite this register only when the counter is stopped (both the TSTART and TCSTF bits of the ULPTCR register are set to 0).

The ULPTISR register sets the polarity of the ULPTEEn pin.

The ULPTISR register is only available for 8-bit read/write access.

25.3.10 ULPTCMSR : ULPT Compare Match Function Select Register

Base address: $ULPTn = 0x4022_0000 + 0x0100 \times n$ ($n = 0, 1$)
 $ULPTn_NS = 0x5022_0000 + 0x0100 \times n$ ($n = 0, 1$)

Offset address: 0x12

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	TOPO LB	TOEB	TCME B	—	TOPO LA	TOEA	TCME A
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TCMEA	Compare Match A Register Enable*1 0: Disable compare match A register 1: Enable compare match A register	R/W
1	TOEA	ULPTOAn Output Enable 0: Disable the ULPTOAn output 1: Enable the ULPTOAn output	R/W
2	TOPOLA	ULPTOAn Polarity Select*2 0: Start the ULPTOAn output when low 1: Start the ULPTOAn output when high	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	TCMEB	Compare Match B Register Enable*1 0: Disable compare match B register 1: Enable compare match B register	R/W
5	TOEB	ULPTOBn Output Enable 0: Disable the ULPTOBn output 1: Enable the ULPTOBn output	R/W
6	TOPOLB	ULPTOBn Polarity Select*2 0: Start the ULPTOBn output when low 1: Start the ULPTOBn output when high	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Do not rewrite this register while the counter is running. Rewrite this register only when the counter is stopped (both the TSTART and TCSTF bits of the ULPTCR register are set to 0).

Note 1. When 1 (stop the counter forcibly) is set to the ULPTCR.TSTOP bit, the TCMEA, and TCMEB bits are set to 0.

Note 2. When data is set to the ULPTMR1 register or 1 is set to the ULPTCR.TSTOP bit, the ULPTOAn and ULPTOBn output is initialized.

The ULPTCMSR register controls the interrupt request signal and input/output of compare match A and compare match B.

The ULPTCMSR register is only available for 8-bit read/write access.

25.4 Operation

25.4.1 Count Operation

The counter is decremented by one on each edge of the count source. When the count value reaches 0x00000000 and the next edge of the count source is input, an interrupt request of underflow is generated. The count operation is set by the ULPTMR1 and the ULPTMR3 registers. [Table 25.4](#) lists the count operations.

Table 25.4 Count operation

Section No.	Count operation			Count source	ULPTMR1 register		ULPTMR3 register	
	Mode 1	Mode 2	Mode 3		TMOD1	TCK1	TCNTCTL	TEECTL[1:0]
section 25.4.1.1. Timer and Continuous Modes	Timer	Continuous	—	ULPTLCLK/ ULPTSCLK	0	0 : ULPTLCLK 1 : ULPTSCLK	0	00
section 25.4.1.5. Common, Continuous, and Count Start Modes			Count start					10
section 25.4.1.6. Common, Continuous, and Count Restart Modes			Count restart					11
section 25.4.1.2. Timer and One-shot Modes		One-shot	—				1	00
section 25.4.1.7. Common, One-shot, and Count Start Modes			Count start					10
section 25.4.1.8. Common, One-shot, and Count Restart Modes			Count restart					11
section 25.4.1.3. Event Counter, Continuous, and Count Enable Modes	Event counter	Continuous	Count enable	ULPTEVIn pin	1	Don't care	0	00
section 25.4.1.5. Common, Continuous, and Count Start Modes			Count start					10
section 25.4.1.6. Common, Continuous, and Count Restart Modes			Count restart					11
section 25.4.1.4. Event Counter, One-shot, and Count Enable Modes		One-shot	Count enable				1	00
section 25.4.1.7. Common, One-shot, and Count Start Modes			Count start					10
section 25.4.1.8. Common, One-shot, and Count Restart Modes			Count restart					11

Note: For details about register settings, see [section 25.3.5. ULPTMR1 : ULPT Mode Register 1](#) and [section 25.3.7. ULPTMR3 : ULPT Mode Register 3](#).

[Mode 1]

The clock used as the count source is selected by the TMOD1 and TCK1 bits of the ULPTMR1 register.

Timer mode: The internal clock (ULPTLCLK or ULPTSCLK) is used as the count source.

The counter is decremented on each rising edge of the count source.

Event counter mode: The external event input (ULPTEVIn pin) is used as the count source.

The counter is decremented on each edge of the count source.

The edge polarity can be selected by the ULPTMR1.TEDGPL bit and the ULPTMR3.TEVPOL bit.

[Mode 2]

The counter behavior is selected by the TCNTCTL bit of the ULPTMR3 register.

Continuous mode: After starting counting, the counter continues to count down (reload value to underflow) repeatedly until the counter stops.

One-shot mode: After starting counting, the counter counts down once (reload value to underflow).

[Mode 3]

The function of the ULPTEEn pin is selected by the TEECTL[1:0] bits of the ULPTMR3 register.

Count enable mode: External events are counted while the ULPTEEn pin is valid.

(This bit is valid only in event counter mode.)

Note: Because the ULPTEEn pin is synchronized with the count source, external events are counted after three cycles of the count source.

Count start mode: The counter starts counting at an edge trigger of the ULPTEEn pin.

The edge polarity of the ULPTEEn pin can be selected by the ULPTMR3.TEEPOL[1:0] bits.

Note: Because the ULPTEEn pin is synchronized and held with the count source, external events are counted after four cycles of the count source.

Count restart mode: The counter starts counting at the first edge trigger of the ULPTEEn pin.

The counter resets the count at the second or following edge trigger.

The edge polarity of the ULPTEEn pin can be selected by the ULPTMR3.TEEPOL[1:0] bits.

Note: Because the ULPTEEn pin is synchronized and held with the count source, external events are counted after four cycles of the count source.

25.4.1.1 Timer and Continuous Modes

This mode operates in timer mode only.

The counter continues to count down (reload value to underflow) repeatedly in synchronization with the count source.

When 1 is set to the ULPTCR.TSTART bit, the counter is started (with the ULPTCR.TCSTF bit set to 1).

When an underflow occurs, the counter counts down again from the reload value.

When 0 is set to the ULPTCR.TSTART bit, the counter is stopped (with the ULPTCR.TCSTF bit set to 0).

Figure 25.2 shows an operation example in timer and continuous modes.

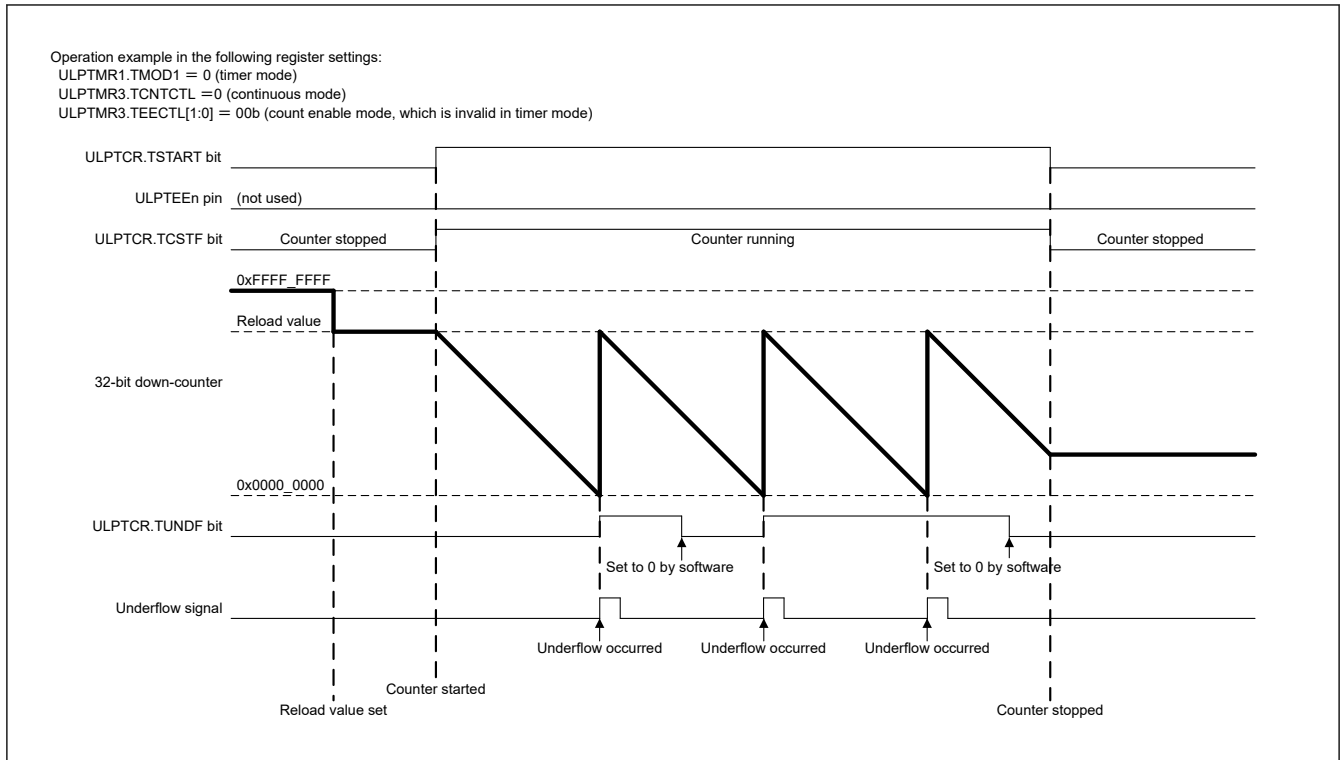


Figure 25.2 Operation example in timer and continuous modes

25.4.1.2 Timer and One-shot Modes

This mode operates in timer mode only.

The counter counts down (reload value to underflow) once in synchronization with the count source.

When 1 is set to the ULPTCR.TSTART bit, the counter is started (with the ULPTCR.TCSTF bit set to 1).

When an underflow occurs, the counter is stopped (with the ULPTCR.TSTART bit set to 1 and the ULPTCR.TCSTF bit set to 0).

To restart the counter after it is stopped, perform the following procedure:

1. Read the ULPTCR.TUNDF bit to confirm that it is set to 1 (underflow occurred).
This is to make sure that counting down is finished.
2. Write 0 to the ULPTCR.TSTART bit to stop the counter.
3. Write 1 to the ULPTCR.TSTART bit to start the counter.

When 0 is set to the ULPTCR.TSTART bit, the counter is stopped (with the ULPTCR.TCSTF bit set to 0).

Figure 25.3 shows an operation example in timer and one-shot modes.

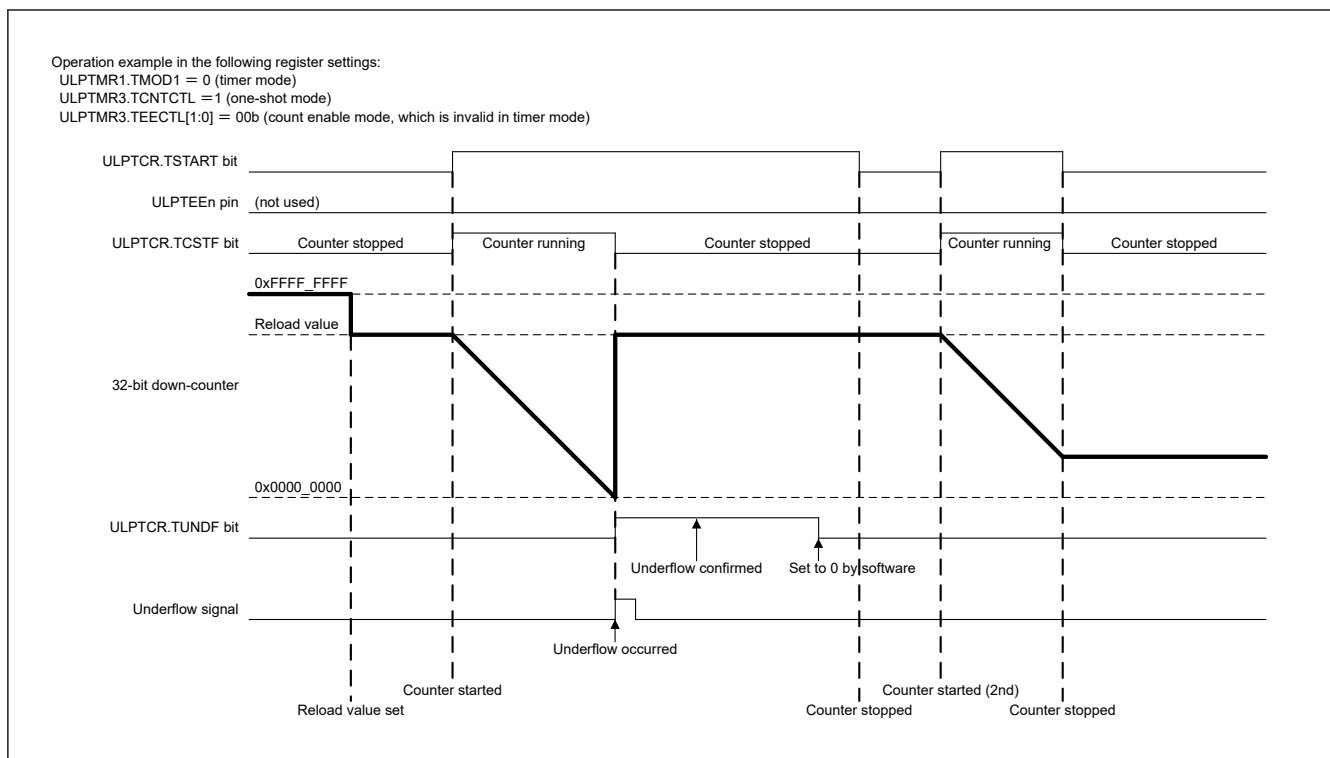


Figure 25.3 Operation example in timer and one-shot modes

25.4.1.3 Event Counter, Continuous, and Count Enable Modes

This mode operates in event counter mode only.

The counter continues to count down (reload value to underflow) repeatedly in synchronization with the count source. The duration in which the count source is counted can be specified by the ULPTIOC.TIOGT0 bit and ULPTISR.RCCPSEL2 bit.

When 1 is set to the ULPTCR.TSTART bit, the counter is started (with the ULPTCR.TCSTF bit set to 1).

When an underflow occurs, the counter counts down again from the reload value.

When 0 is set to the ULPTCR.TSTART bit, the counter is stopped (with the ULPTCR.TCSTF bit set to 0).

Figure 25.4 shows an operation example when the ULPTIOC.TIOGT0 bit is set to 0. Figure 25.5 and Figure 25.6 show operation examples when the ULPTIOC.TIOGT0 bit is set to 1.

Operation example in the following register settings:
 ULPTMR1.TMOD1 = 1 (event counter mode)
 ULPTMR3.TCNTCTL = 0 (continuous mode)
 ULPTMR3.TEECTL[1:0] = 00b (count enable mode)
 ULPTIOC.TIOGT0 = 0 (external events are always counted)
 ULPTISR.RCCPSEL2 = don't care

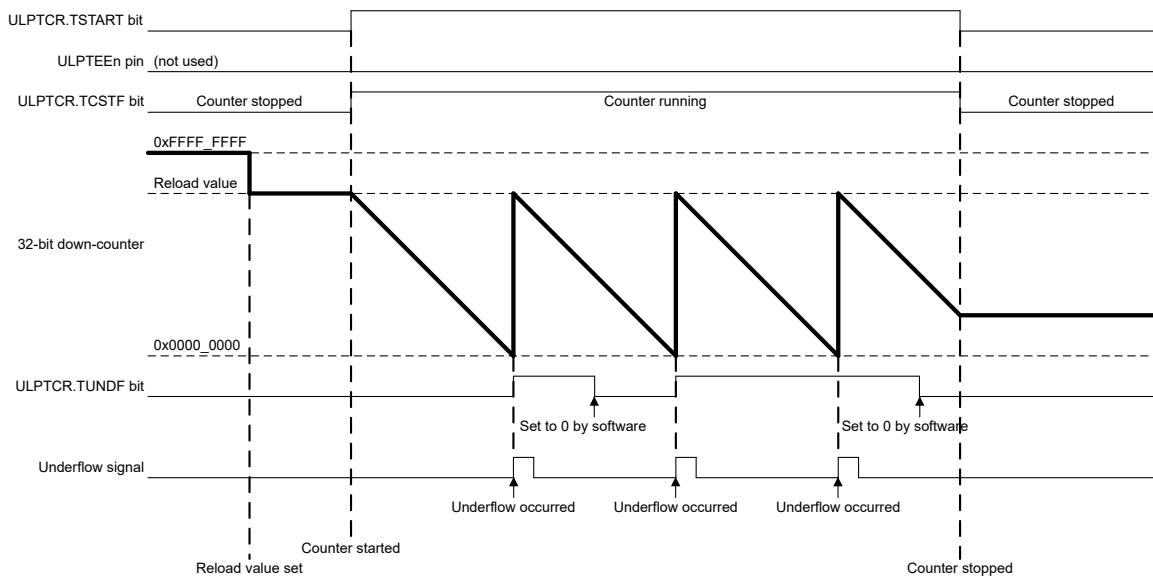


Figure 25.4 Operation example in event counter, continuous, and count enable modes (with the ULPTIOC.TIOGT0 bit set to 0)

Operation example in the following register settings:
 ULPTMR1.TMOD1 = 1 (event counter mode)
 ULPTMR3.TCNTCTL = 0 (continuous mode)
 ULPTMR3.TEECTL[1:0] = 00b (count enable mode)
 ULPTIOC.TIOGT0 = 1 (external events are counted while ULPTEn is valid)
 ULPTISR.RCCPSEL2 = 0 (external events are counted while low)

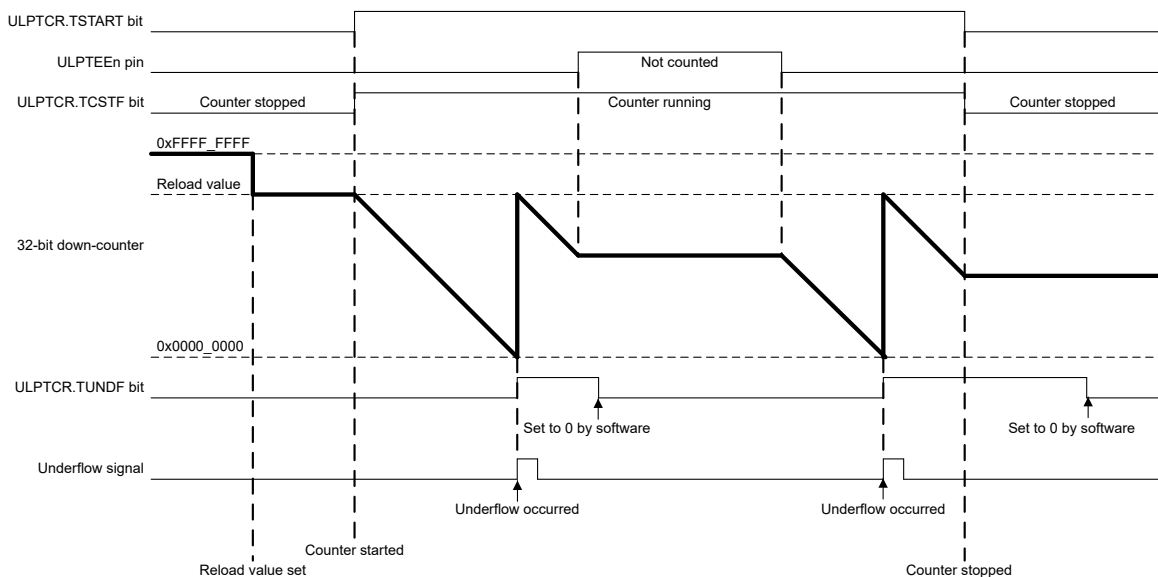
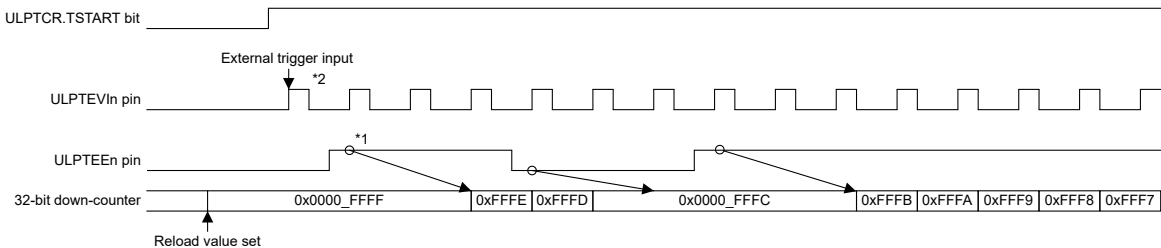


Figure 25.5 Operation example 1 in event counter, continuous, and count enable modes (with the ULPTIOC.TIOGT0 bit set to 1)

Operation example in the following register settings:

ULPTMR1.TMOD1 = 1 (event counter mode)
 ULPTMR3.TCNTCTL = 0 (continuous mode)
 ULPTMR3.TEECTL[1:0] = 00b (count enable mode)
 ULPTIOC.TIOGT0 = 1 (external events are counted while ULPTEEn is valid)
 ULPTISR.RCCPSEL2 = 1 (external events are counted while high)
 ULPTMR3.TEVPOL = 0 (ULPTEVin is inverted)



- Note 1. When synchronization is controlled, there is a delay by two cycles of count source until external events are counted. In addition, the count start timing may be shifted by one cycle due to the phase difference between the ULPTEEn pin and the sampling clock.
- Note 2. Depending on the state before the counter is stopped, counting may be performed during two cycles of the count source immediately after the counter is started.
 To disable the counting operation during two cycles immediately after the counter is started, set 1 to the ULPTCR.TSTOP bit to initialize the internal circuits, and then finish the operation settings before starting the counter.

Figure 25.6 Operation example 2 in event counter, continuous, and count enable modes (with the ULPTIOC.TIOGT0 bit set to 1)

25.4.1.4 Event Counter, One-shot, and Count Enable Modes

This mode operates in event counter mode only.

The counter counts down (reload value to underflow) once in synchronization with the count source. The duration in which the count source is counted can be specified by the ULPTIOC.TIOGT0 bit and ULPTISR.RCCPSEL2 bit.

When 1 is set to the ULPTCR.TSTART bit, the counter is started (with the ULPTCR.TCSTF bit set to 1).

When an underflow occurs, the counter is stopped (with the ULPTCR.TSTART bit set to 1 and the ULPTCR.TCSTF bit set to 0).

To restart the counter after the counter is stopped, perform the following procedure:

1. Read the ULPTCR.TUNDF bit to confirm that it is set to 1 (underflow occurred).
This is to make sure that counting down is finished.
2. Write 0 to the ULPTCR.TSTART bit to stop the counter.
3. Write 1 to the ULPTCR.TSTART bit to start the counter.

When 0 is set to the ULPTCR.TSTART bit, the counter is stopped (with the ULPTCR.TCSTF bit set to 0).

Figure 25.7 shows an operation example when the ULPTIOC.TIOGT0 bit is set to 00, and Figure 25.8 shows an operation example when the ULPTIOC.TIOGT0 bit is set to 01.

Operation example in the following register settings:
 ULPTMR1.TMOD1 = 1 (event counter mode)
 ULPTMR3.TCNTCTL = 1 (one-shot mode)
 ULPTMR3.TEECTL[1:0] = 00b (count enable mode)
 ULPTIOC.TIOGT0 = 0 (external events are always counted)
 ULPTISR.RCCPSEL2 = don't care

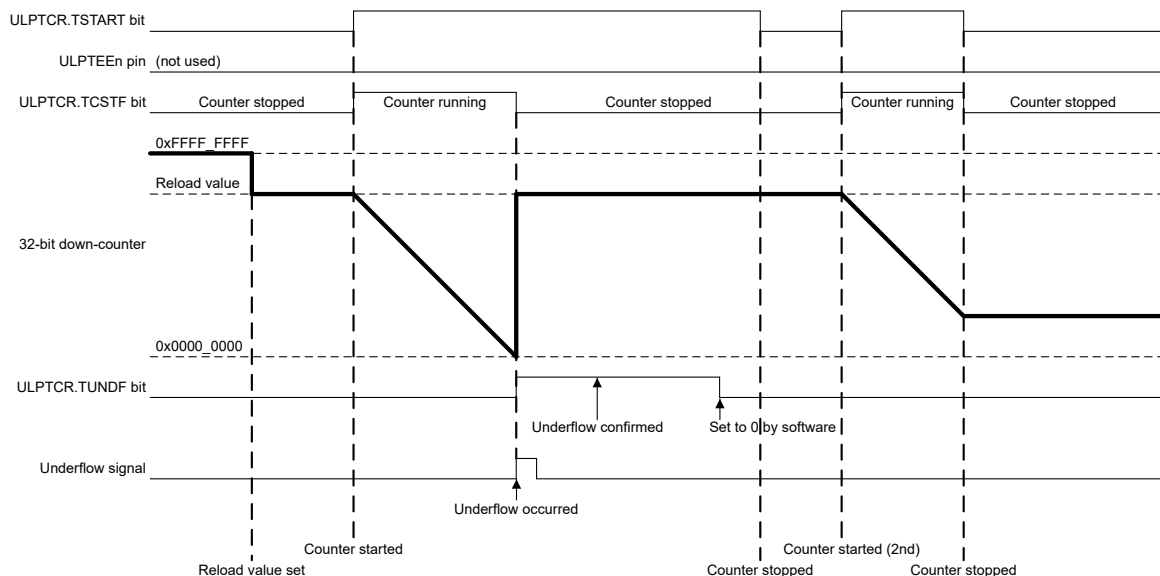


Figure 25.7 Operation example in event counter, one-shot, and count enable modes (with the ULPTIOC.TIOGT0 bit set to 0)

Operation example in the following register settings:
 ULPTMR1.TMOD1 = 1 (event counter mode)
 ULPTMR3.TCNTCTL = 1 (one-shot mode)
 ULPTMR3.TEECTL[1:0] = 00b (count enable mode)
 ULPTIOC.TIOGT0 = 1 (external events are counted while ULPTEn is valid)
 ULPTISR.RCCPSEL2 = 0 (external events are counted while low)

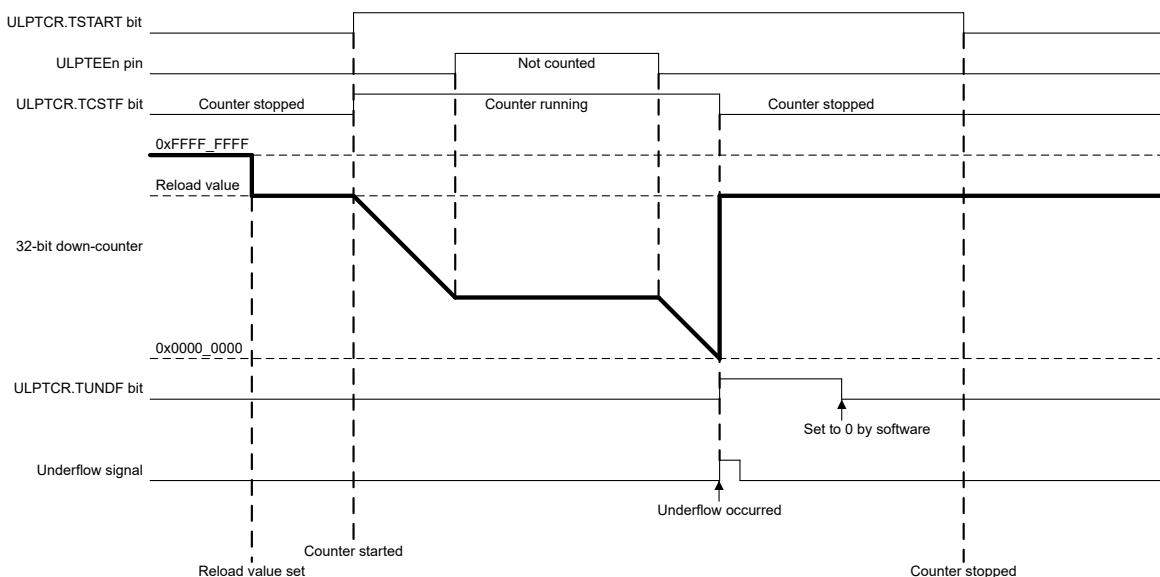


Figure 25.8 Operation example in event counter, one-shot, and count enable modes (with the ULPTIOC.TIOGT0 bit set to 1)

25.4.1.5 Common, Continuous, and Count Start Modes

This mode operates in both timer mode and event counter mode.

The counter continues to count down (reload value to underflow) repeatedly in synchronization with the count source.

After 1 is set to the ULPTCR.TSTART bit, the counter is started (with the ULPTCR.TCSTF bit set to 1) by an edge trigger at the ULPTEEn pin.

When an underflow occurs, the counter counts down again from the reload value.

When 0 is set to the ULPTCR.TSTART bit, the counter is stopped (with the ULPTCR.TCSTF bit set to 0).

Figure 25.9 shows an operation example in common, continuous, and count start modes.

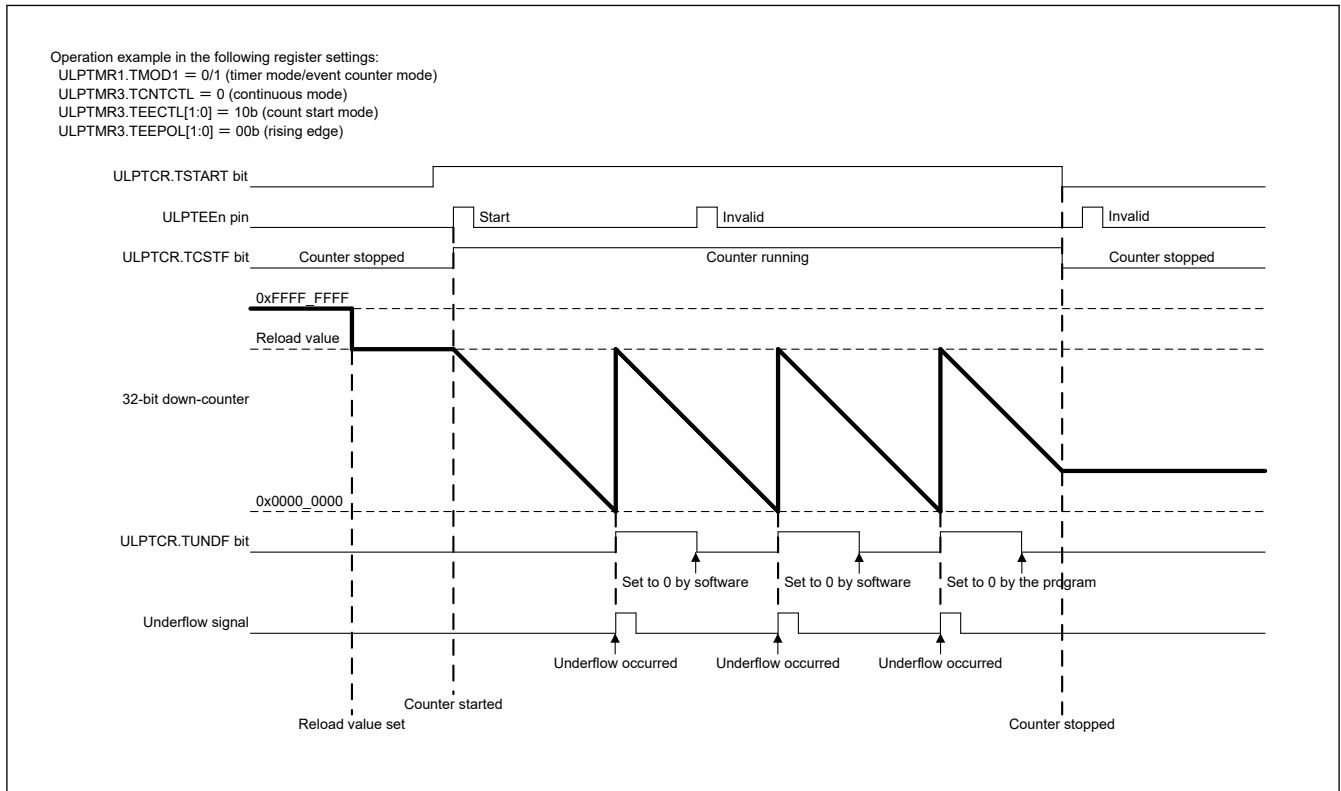


Figure 25.9 Operation example in common, continuous, and count start modes

25.4.1.6 Common, Continuous, and Count Restart Modes

This mode operates in both timer mode and event counter mode.

The counter continues to count down (reload value to underflow) repeatedly in synchronization with the count source.

After 1 is set to the ULPTCR.TSTART bit, the counter is started (with the ULPTCR.TCSTF bit set to 1) by an edge trigger at the ULPTEEn pin.

On an edge trigger input at the ULPTEEn pin when the counter is running (with the ULPTCR.TCSTF bit set to 1), the counter is reset and it starts counting again from the reload value.

If no edge trigger is input from the ULPTEEn pin and an overflow occurs, the counter starts counting again from the reload value.

When 0 is set to the ULPTCR.TSTART bit, the counter is stopped (with the ULPTCR.TCSTF bit set to 0).

Figure 25.10 shows an operation example in common, continuous, and count restart modes.

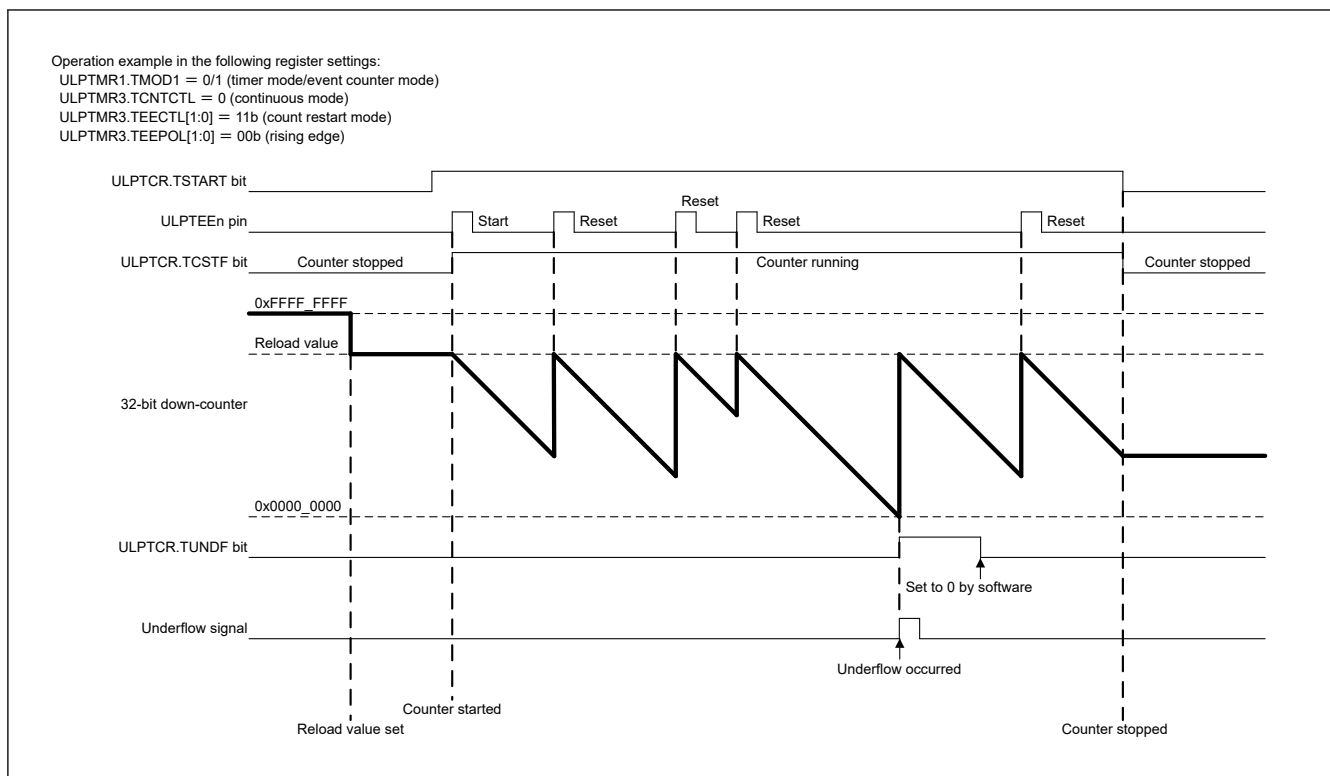


Figure 25.10 Operation example in common, continuous, and count restart modes

25.4.1.7 Common, One-shot, and Count Start Modes

This mode operates in both timer mode and event counter mode.

The counter counts down (reload value to underflow) once in synchronization with the count source.

After 1 is set to the ULPTCR.TSTART bit, the counter is started (with the ULPTCR.TCSTF bit set to 1) by an edge trigger at the ULPTEEn pin.

When an underflow occurs, the counter is stopped (with the ULPTCR.TSTART bit set to 1 and the ULPTCR.TCSTF bit set to 0).

On an edge trigger input at the ULPTEEn pin when the counter is running (with ULPTCR.TSTART bit set to 1 and the ULPTCR.TCSTF bit set to 0), the counter starts counting down again from the reload value.

When 0 is set to the ULPTCR.TSTART bit, the counter is stopped (with the ULPTCR.TCSTF bit set to 0).

Figure 25.11 shows an operation example in common, one-shot, and count start modes.

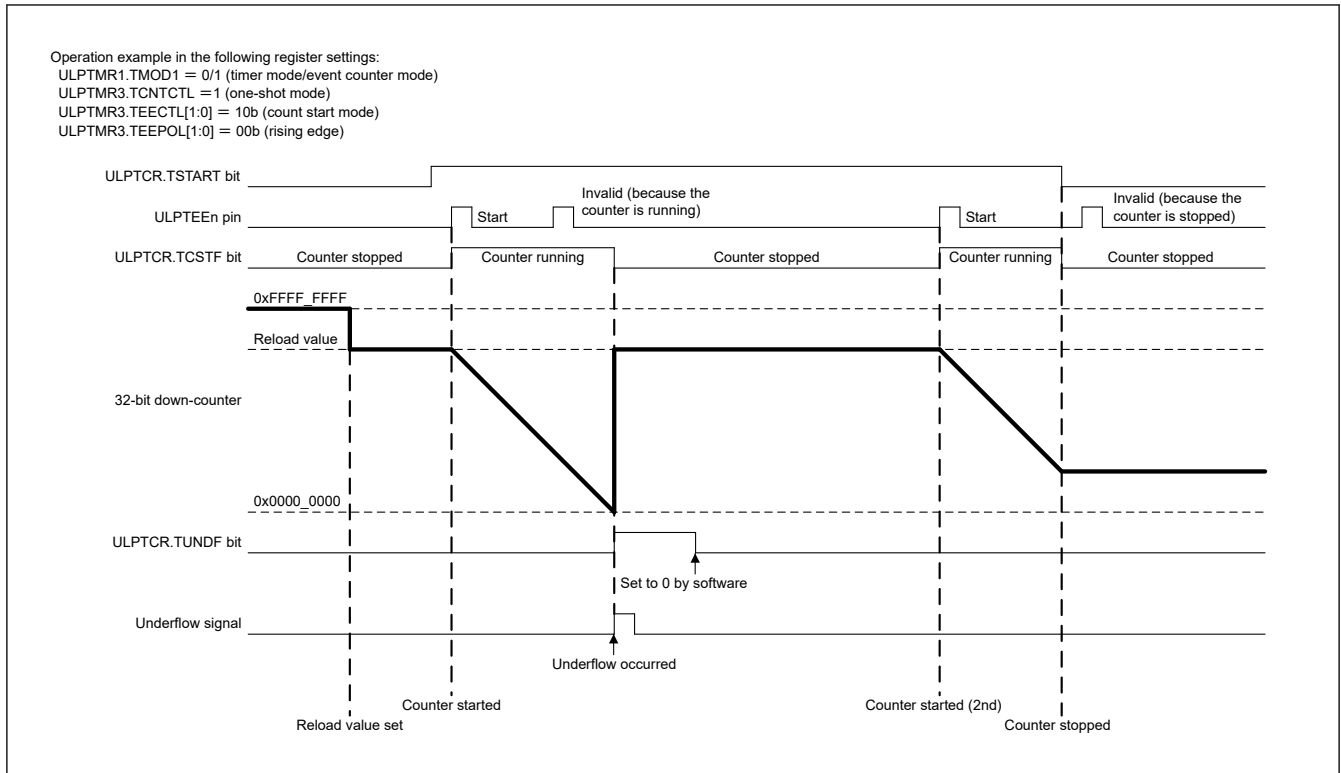


Figure 25.11 Operation example in common, one-shot, and count start modes

25.4.1.8 Common, One-shot, and Count Restart Modes

This mode operates in both timer mode and event counter mode.

The counter counts down (reload value to underflow) once in synchronization with the count source.

After 1 is set to the ULPTCR.TSTART bit, the counter is started (with the ULPTCR.TCSTF bit set to 1) by an edge trigger of the ULPTEEn pin.

On an edge trigger input at the ULPTEEn pin when the counter is running (with the ULPTCR.TCSTF bit set to 1), the counter is reset and it starts counting again from the reload value.

If no edge trigger is input at the ULPTEEn pin and an underflow occurs, the counter is stopped (with the ULPTCR.TSTART bit set to 1 and the ULPTCR.TCSTF bit set to 0).

On an edge trigger input at the ULPTEEn pin when the counter is running (with the ULPTCR.TSTART bit set to 1 and the ULPTCR.TCSTF bit set to 0), the counter starts counting down again from the reload value.

When 0 is set to the ULPTCR.TSTART bit, the counter is stopped (with the ULPTCR.TCSTF bit set to 0).

Figure 25.12 shows an operation example in common, one-shot, and count restart modes.

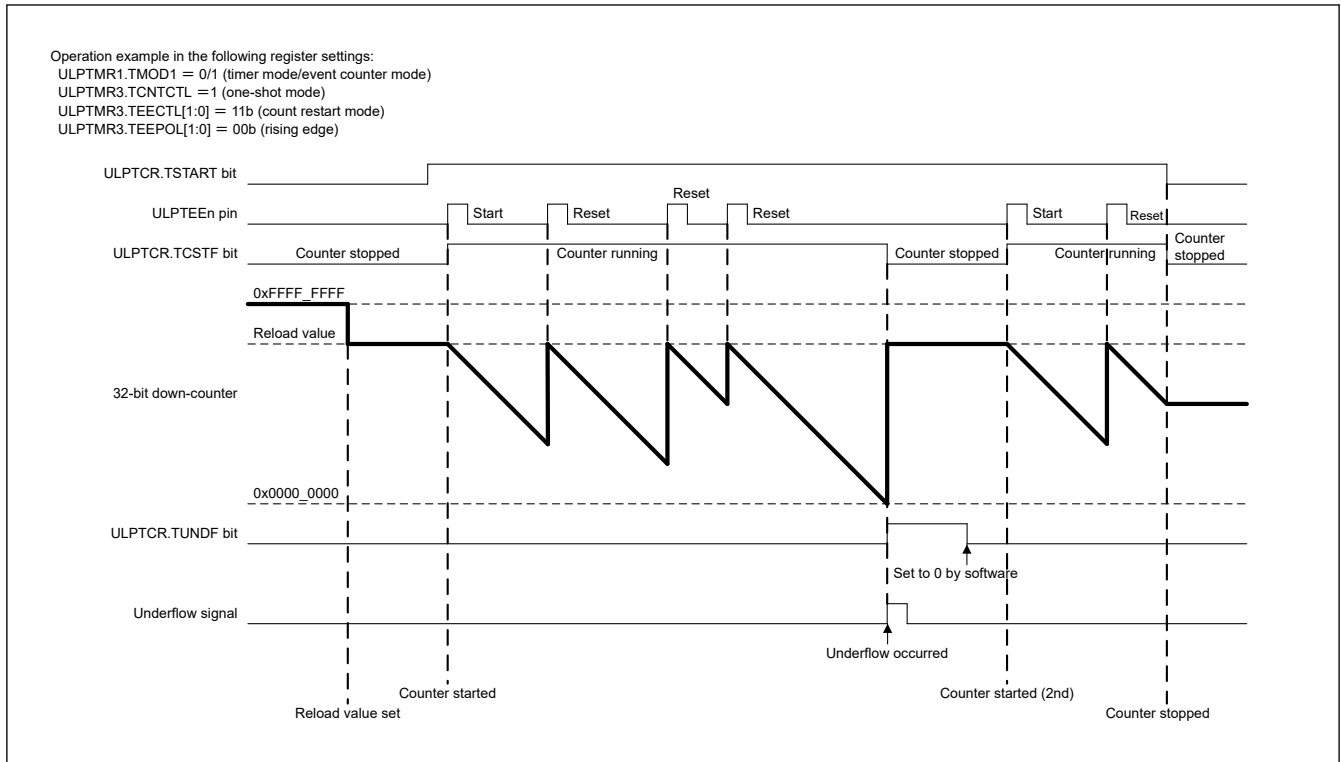


Figure 25.12 Operation example in common, one-shot, and count restart modes

25.4.2 Rewriting the Counter and Reload Register

Regardless of the operating mode, the counter and reload register rewrite timing changes depending on the following bits:

- TSTART and TCSTF bits of the ULPTCR register
- TCMEA and TCMEB bits of the ULPTCMSR register
- TEECTL[1:0] bits of the ULPTMR3 register.

Condition A: When the counter is stopped (both the TSTART and TCSTF bits of the ULPTCR register are set to 0)

A reload value is written directly to the counter and the reload register.

Condition B: When the counter is running (with the ULPTCR.TCSTF bit set to 1) and the compare match A/B registers are disabled (with both the TCMEA and TCMEB bits of the ULPTCR register set to 0)

After a reload value is written to the ULPTCNT register, the reload value is first set to the reload register, then to the counter in synchronization with the count source.

Condition C: When the counter is running (with the ULPTCR.TCSTF bit set to 1) and the compare match A/B registers are enabled (with either the TCMEA bit or the TCMEB bit of the ULPTCR register set to 1)

After a reload value is written to the ULPTCNT register, the reload value is written to the reload register in synchronization with the count source. Then, the reload value is set to the counter in synchronization with an underflow.

Condition D: When the counter is running (with ULPTCR.TCSTF bit set to 1) and when the count restart mode is enabled (with the ULPTMR3.TEECTL[1:0] bits set to 11)

After an edge trigger is input from the ULPTEEn pin, the reload value set in the reload register is set to the counter in synchronization with the count source. In this condition, the reload register is not rewritten.

Figure 25.13 to Figure 25.15 show the counter and reload register rewrite timing.

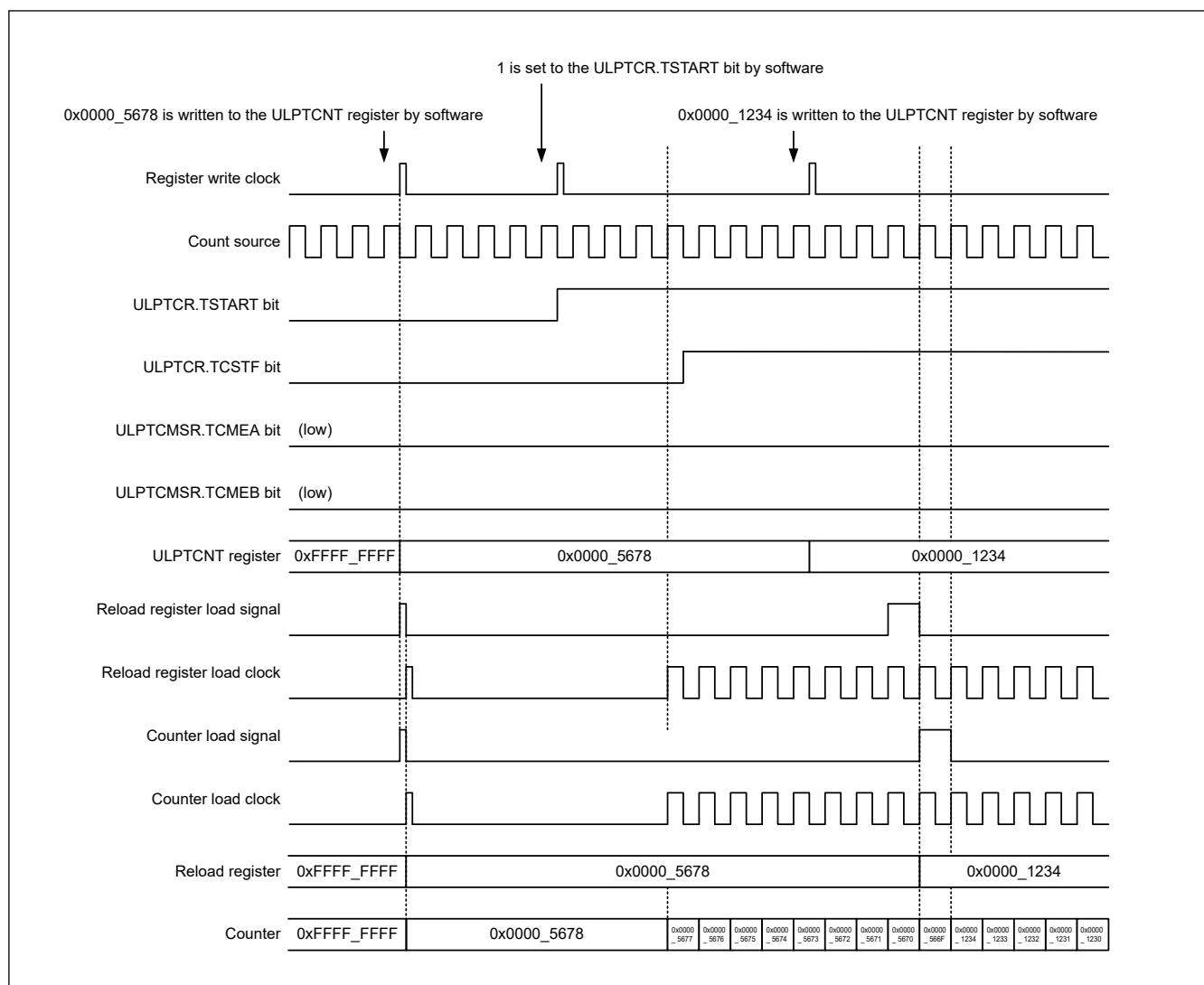


Figure 25.13 Counter and reload rewrite timing (under condition A or B)

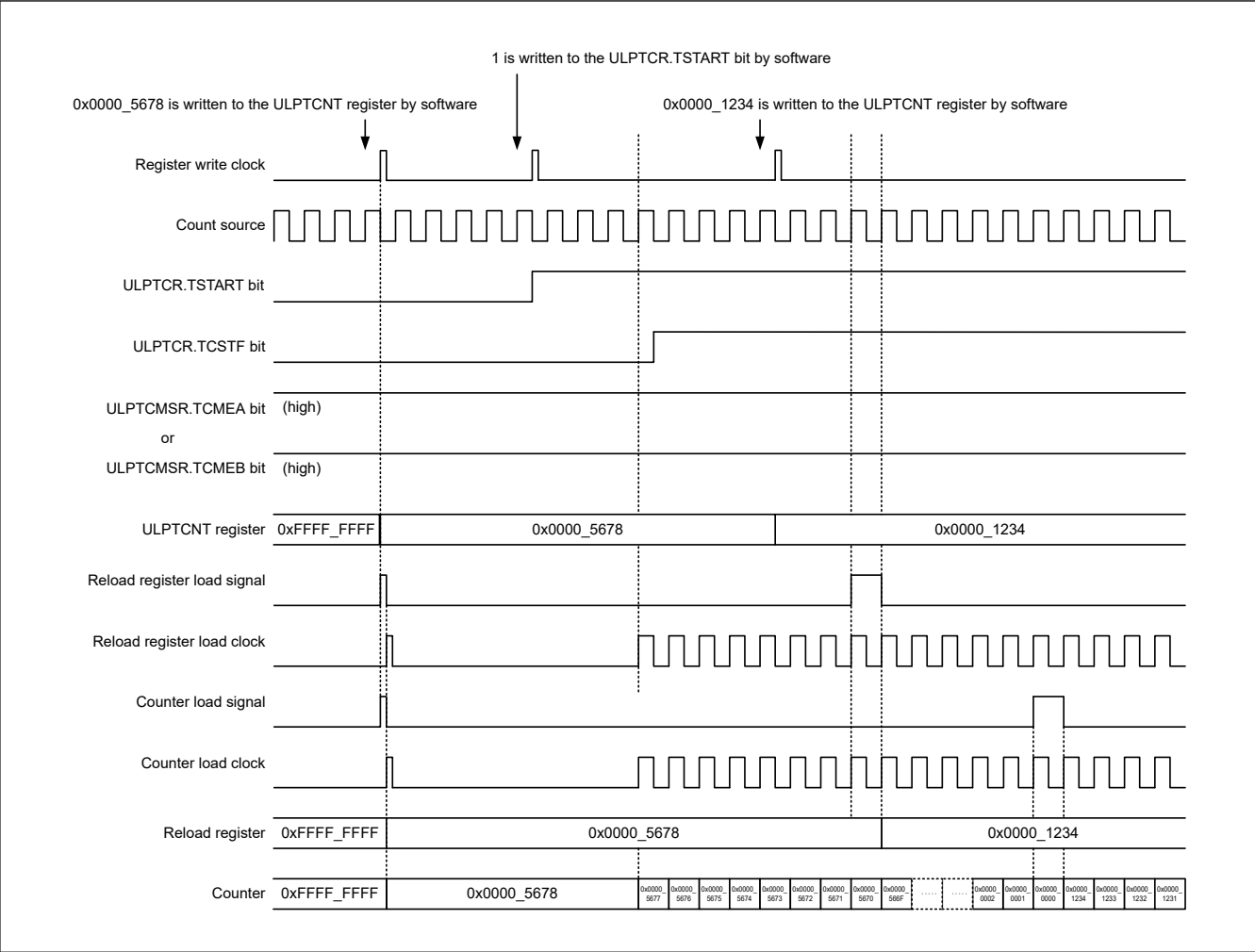


Figure 25.14 Counter and reload rewrite timing (under condition A or C)

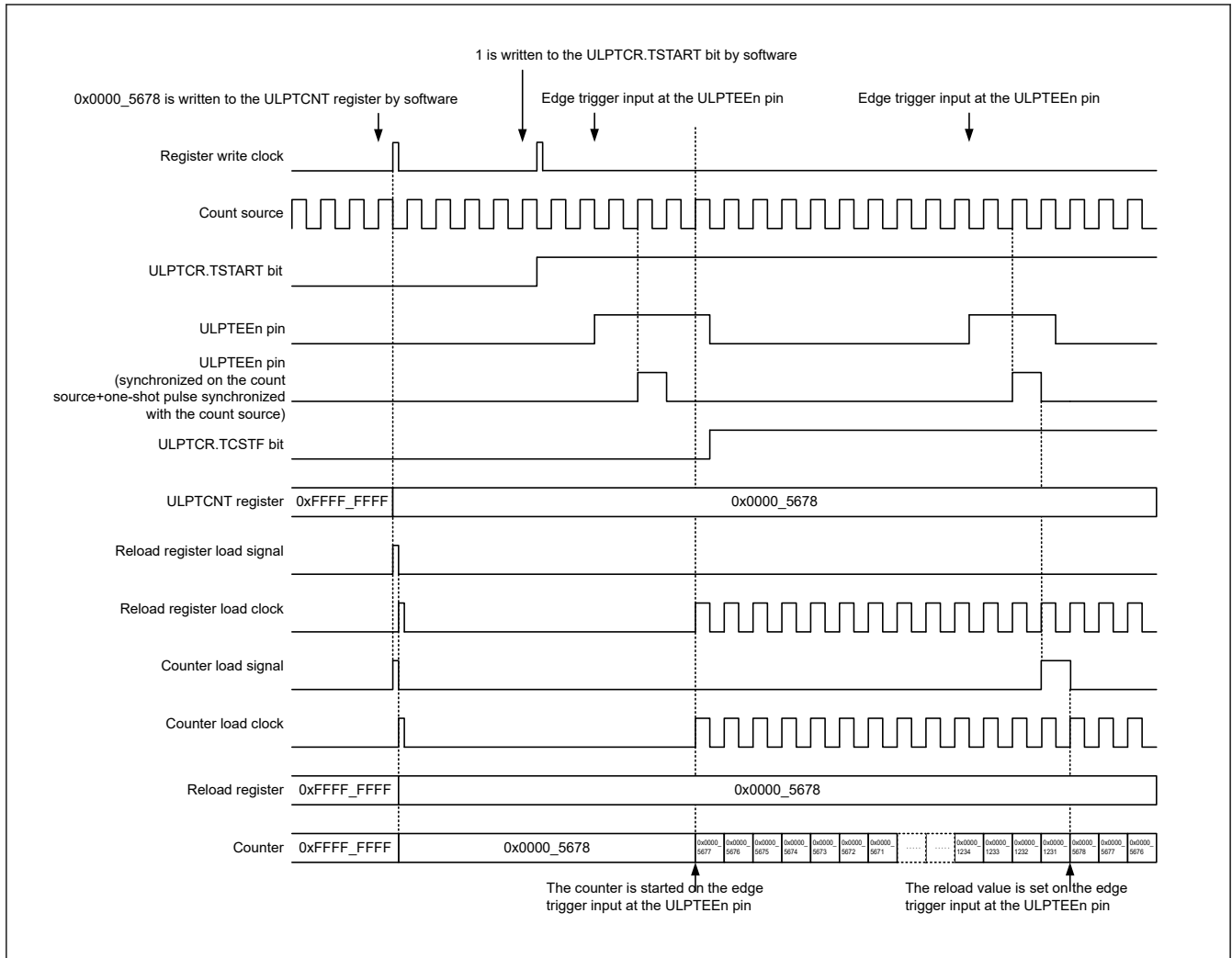


Figure 25.15 Counter and reload rewrite timing (under condition D)

25.4.3 Rewriting the Compare Circuits and Reload Registers for Compare Match A/B

Regardless of the operating mode, the rewrite timing for the compare circuits and reload registers for compare match A/B changes depending on the `TSTART` and `TCSTF` bits of the `ULPTCR` register.

Condition A: When the counter is stopped (both the `TSTART` and `TCSTF` bits of the `ULPTCR` register are set to 0)

Compare match A/B values are written directly to the compare circuits and reload registers.

Condition B: When the counter is running (with the `ULPTCR.TCSTF` bit set to 1)

After compare match A/B values are written to the `ULPTCMA` and `ULPTCMB` registers, the compare match A/B values are written to the reload registers in synchronization with the count source. Then, the compare match A/B values are written to the compare circuits in synchronization with an underflow.

Figure 25.16 shows the rewrite timing for the compare circuit and reload register for compare match A.

The timing for compare match B is similar to that for compare match A.

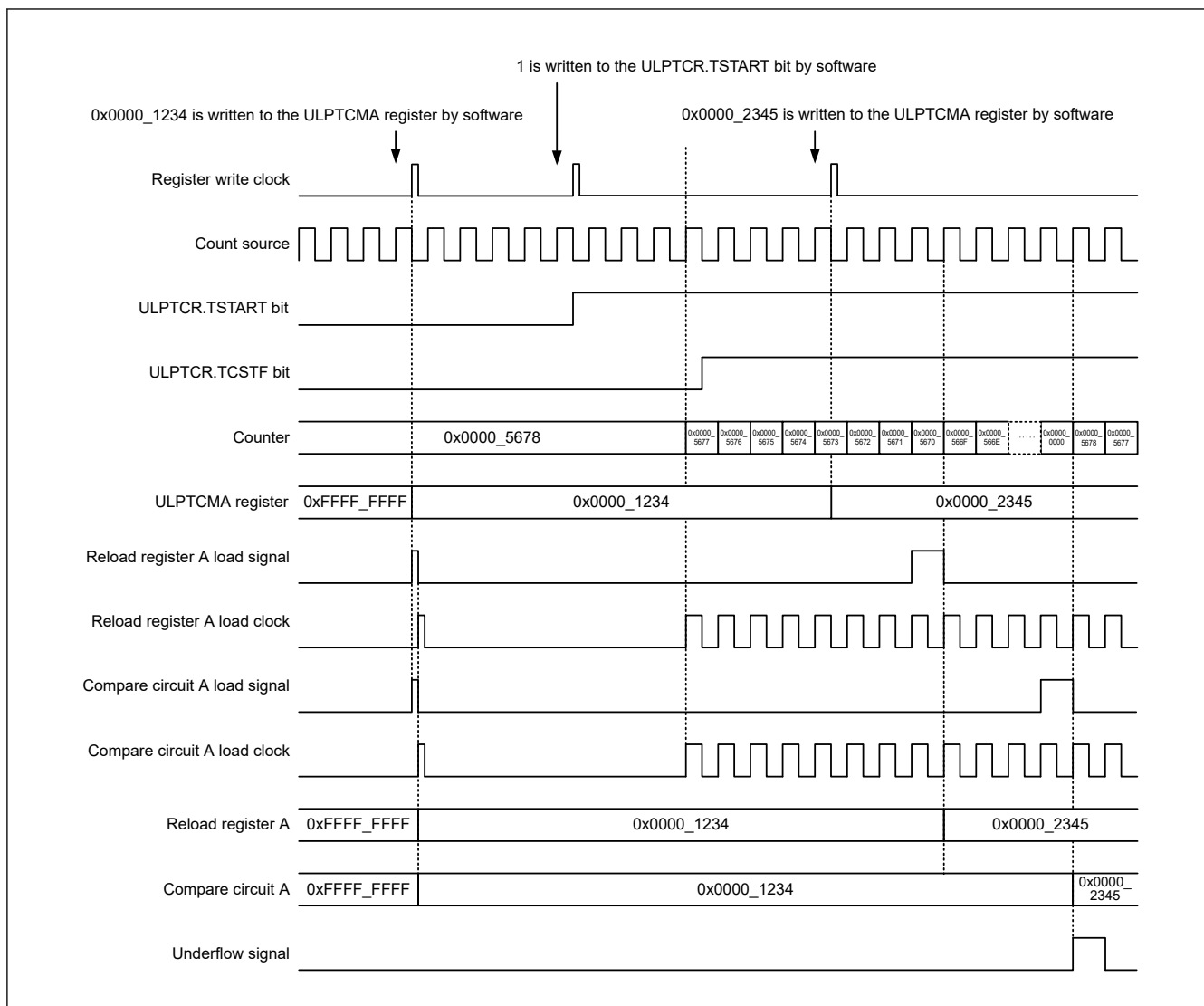


Figure 25.16 Rewrite timing for the compare circuit and reload register for compare match A

25.4.4 Pulse Output

Regardless of the operating mode, the ULPTOn pin can output pulses. The output level is inverted each time the counter underflows. The pulse output from the ULPTOn pin can be stopped by the ULPTIOC.TOE bit. The initial value of the output level can be selected by the ULPTMR3.TOPOL bit.

Figure 25.17 shows an operation example of a pulse output.

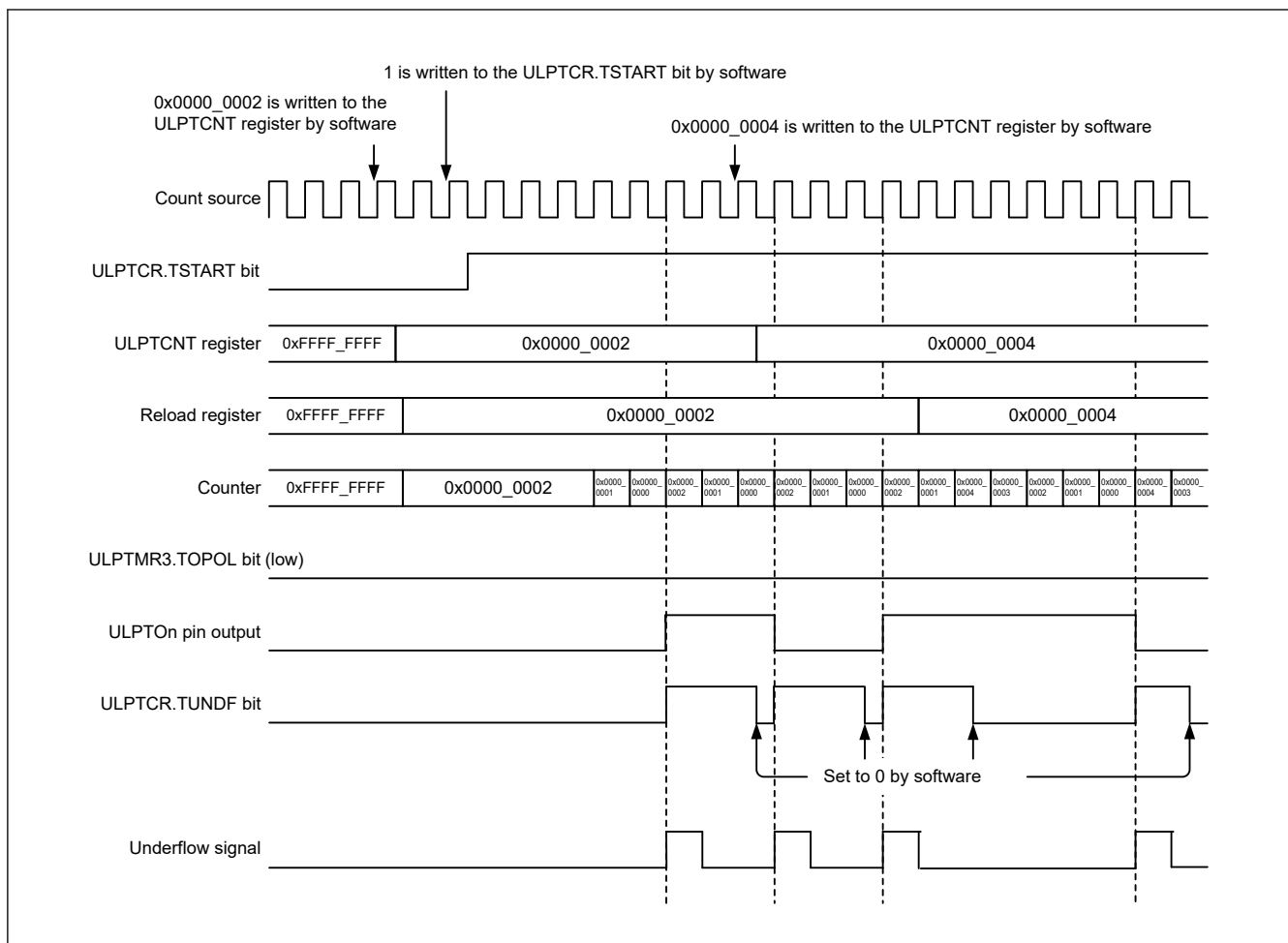


Figure 25.17 Operation example of pulse output

25.4.5 Compare Match Function

This function detects a match (compare match) between the ULPTCMA/ULPTCMB register contents and the ULPTCNT register contents. This function is enabled when the TCMEA or TCMEB bit of the ULPTCMSR register is set to 1 (the compare match A/B registers are enabled).

The counter is decremented in synchronization with the count source selected by the TMOD1/TCK1 bits of the ULPTMR1 register. When the ULPTCNT register value matches the ULPTCMA or ULPTCMB register value, the TCMAF or TCMBF bit of the ULPTCR register is set to 1 and an interrupt request is generated.

The counter and reload register rewrite timing differs depending on whether the compare match function is enabled. For details, see [section 25.4.2. Rewriting the Counter and Reload Register](#).

The output level of the ULPTOAn/ULPTOBn pin is inverted on a match or underflow. The output level can be selected by the ULPTOAn/ULPTOBn bit of the ULPTCMSR register.

[Figure 25.18](#) shows an operation example of the compare match function.

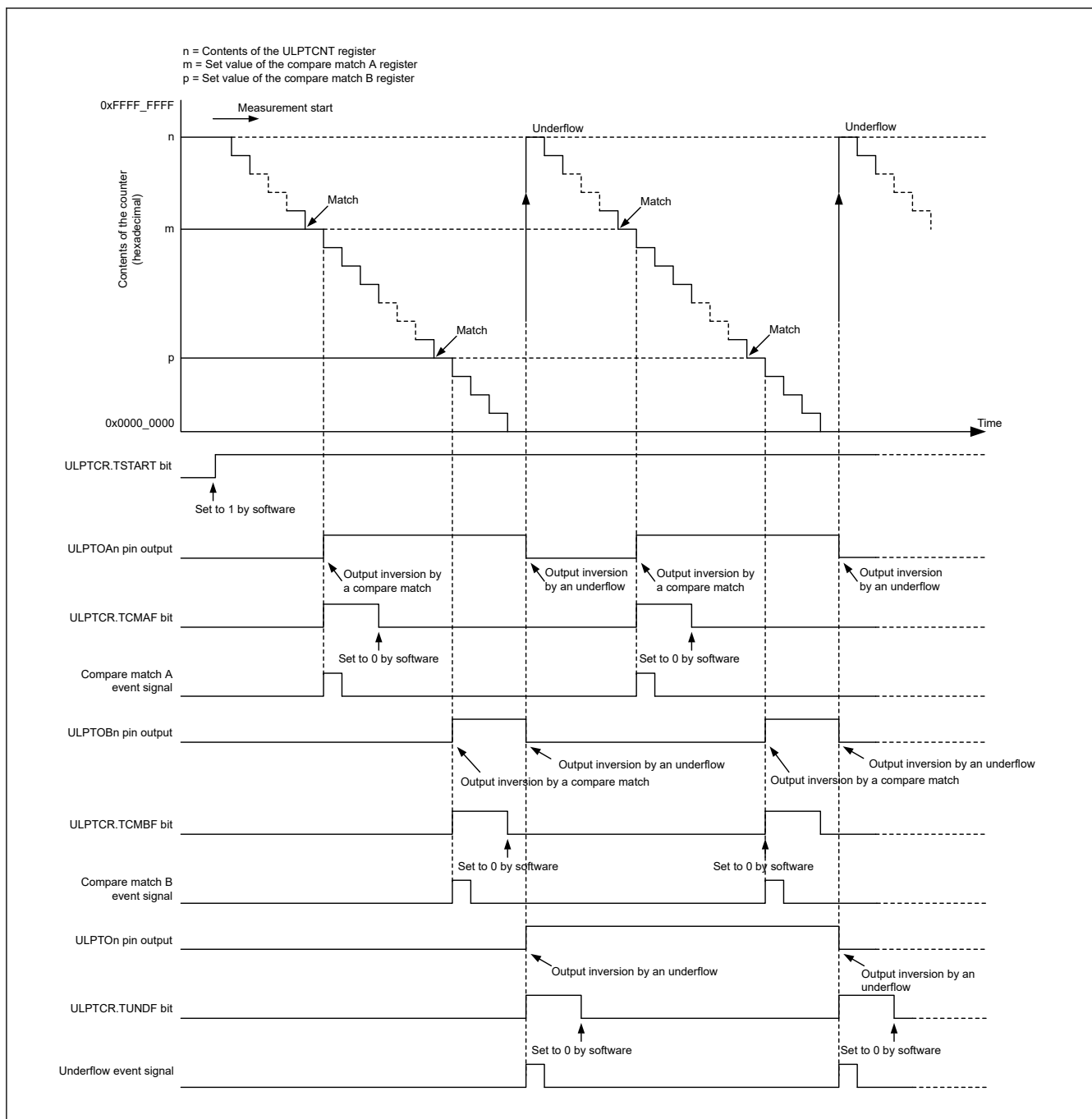


Figure 25.18 Operation example of the compare match function (both the TOPOLA and TOPOLB bits of the ULPTCMSR register are set to 0)

25.4.6 Input and Output Settings for each Mode

Table 25.5 to Table 25.9 show the states of the input pins (ULPTEEn and ULPTEVIn) and output pins (ULPTOn, ULPTOAn, and ULPTOBn) in each mode.

Table 25.5 ULPTEEn pin settings

Operating mode		ULPTMR1 register	ULPTMR3 register	ULPTEEn pin input
		TMOD1 bit	TEECTL[1:0] bit	
Timer mode	—	0	00	Input disabled
	Count start		10	Normal input
	Count restart		11	
Event counter mode	Count enable	1	00	
	Count start		10	
	Count restart		11	

Table 25.6 ULPTEVIn pin settings

Operating mode	ULPTMR1 register	ULPTMR3 register	ULPTEVIn pin input
	TMOD1 bit	TEVPOL bit	
Timer mode	0	0 or 1	Input disabled
Event counter mode	1	0	Normal input
		1	Inverted input

Table 25.7 ULPTOn pin settings

Operating mode	ULPTIOC register	ULPTMR3 register	ULPTOn pin output
	TOE bit	TOPOL bit	
All modes	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled

Table 25.8 ULPTOAn pin settings

Operating mode	ULPTCMSR register		ULPTOAn pin output
	TOEA bit	TOPOLA bit	
All modes	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled

Table 25.9 ULPTOBn pin settings

Operating mode	ULPTCMSR register		ULPTOBn pin output
	TOEB bit	TOPOLB bit	
All modes	1	1	Inverted output
		0	Normal output
	0	0 or 1	Output disabled

25.4.7 Standby Mode

The ULPT can operate in Software Standby mode or Deep Software Standby mode 1. Set it to each standby mode with count operation start (ULPTCR.TSTART = 1, ULPTCR.TCSTF = 1).

Table 25.10 lists the settings available in Software Standby mode or Deep Software Standby mode 1.

Table 25.10 Settings available in Software Standby mode or Deep Software Standby mode 1

Section No.	Count operation			Count source	CPU recovery source
	Mode 1	Mode 2	Mode 3		
section 25.4.1.1. Timer and Continuous Modes	Timer	Continuous	—	ULPTLCLK, ULPTSCLK	- Underflow - Compare match A/B
section 25.4.1.5. Common, Continuous, and Count Start Modes			Count start		
section 25.4.1.6. Common, Continuous, and Count Restart Modes			Count restart		
section 25.4.1.2. Timer and One-shot Modes		One-shot	—		
section 25.4.1.7. Common, One-shot, and Count Start Modes			Count start		
section 25.4.1.8. Common, One-shot, and Count Restart Modes			Count restart		
section 25.4.1.3. Event Counter, Continuous, and Count Enable Modes	Event counter	Continuous	Count enable	ULPTEVIn pin ^{*1}	- Underflow - Compare match A/B
section 25.4.1.5. Common, Continuous, and Count Start Modes			Count start		
section 25.4.1.6. Common, Continuous, and Count Restart Modes			Count restart		
section 25.4.1.4. Event Counter, One-shot, and Count Enable Modes		One-shot	Count enable		
section 25.4.1.7. Common, One-shot, and Count Start Modes			Count start		
section 25.4.1.8. Common, One-shot, and Count Restart Modes			Count restart		

Note: The ULPT can operate in all modes even in Software Standby mode or Deep Software Standby mode 1. For details about the count operations and register settings, see [section 25.4.1. Count Operation](#).

Note 1. Only ULPTEVIn pin can be used in Deep Software Standby mode 1.

Only in event counter mode (ULPTMR1.TMOD1 = 1), set the ULPTIOC.TIPF[1:0] bits to 00 (no filter) before entering standby mode.

In Software Standby mode or Deep Software Standby mode 1, the digital filter clock (PCLKB) stops (the digital filter does not function).

It is prohibited to rewrite the ULPTCNT, ULPTCMA, and ULPTCMB registers immediately before setting each standby mode.

If the ULPTCNT, ULPTCMA, and ULPTCMB registers are rewritten while the counter is running, set each standby mode after four or more cycles of the count source.

25.4.8 Interrupt Sources

The ULPT has three interrupt sources:

- ULPTI
- ULPTCMAI
- ULPTCMBI.

[Table 25.11](#) lists the interrupt sources.

Table 25.11 ULPT interrupt sources (1 of 2)

Name	Interrupt source	DMAC/DTC activation
ULPTn_ULPTI	When the counter underflows	Possible
ULPTn_ULPTCMAI	When the ULPTCNT value matches the ULPTCMA value	Possible

Table 25.11 ULPT interrupt sources (2 of 2)

Name	Interrupt source	DMAC/DTC activation
ULPTn_ULPTCMBI	When the ULPTCNT value matches the ULPTCMB value	Possible

25.4.9 Event Output to the ELC

The ULPT is capable of link operation for a specified module through the event link controller (ELC) using the interrupt request signal as an event signal.

The ULPT outputs an event signal when compare match A/compare match B or an underflow occurs. For details, see [section 19, Event Link Controller \(ELC\)](#).

25.5 Usage Notes

25.5.1 Start and Stop Control of the Counter

Condition A: In timer mode (ULPTMR1.TMOD1 = 0), continuous mode (ULPTMR3.TCNTCTL = 0), and count enable mode (ULPTMR3.TEECTL[1:0] = 00)

When the counter is stopped (with the TCSTF bit of the ULPTCR register set to 0), even if 1 (start the counter) is set to the TSTART bit of the ULPTCR register, the TCSTF bit remains at 0 (counter stopped) for five cycles of the count source.

Do not access the registers^{*1} related to the ULPT other than the TCSTF bit until the TCSTF bit is set to 1 (counter running).

When the counter is running (with the TCSTF bit set to 1), even if 0 (stop the counter) is set to the TSTART bit, the TCSTF bit remains at 1 (counter running) for five cycles of the count source.

Do not access the registers^{*1} related to the ULPT other than the TCSTF bit until the TCSTF bit is set to 0 (counter stopped).

Before changing the TSTART bit from 0 to 1, clear the interrupt register. For details, see [section 14, Interrupt Controller Unit \(ICU\)](#).

Note 1. Registers related to the ULPT: ULPTCNT, ULPTCMA, ULPTCMB, ULPTCR, ULPTMR1, ULPTMR2, ULPTMR3, ULPTIOC, ULPTISR, and ULPTCMSR.

Condition B: In event counter mode (ULPTMR1.TMOD1 = 1), continuous mode (ULPTMR3.TCNTCTL = 0), and count enable mode (ULPTMR3.TEECTL[1:0] = 00)

When the counter is stopped (with the TCSTF bit of the ULPTCR register set to 0), even if 1 (start the counter) is set to the TSTART bit of the ULPTCR register, the TCSTF bit remains at 0 (counter stopped) for two cycles of the count source.

Do not access the registers^{*1} related to the ULPT other than the TCSTF bit until the TCSTF bit is set to 1 (counter running).

When the counter is running (with the TCSTF bit set to 1), even if 0 (stop the counter) is set to the TSTART bit, the TCSTF bit remains at 1 (counter running) for two cycles of the count source.

Do not access the registers^{*1} related to the ULPT other than the TCSTF bit until the TCSTF bit is set to 0 (counter stopped).

Before changing the TSTART bit from 0 to 1, clear the interrupt register. For details, see [section 14, Interrupt Controller Unit \(ICU\)](#).

Note 1. Registers related to the ULPT: ULPTCNT, ULPTCMA, ULPTCMB, ULPTCR, ULPTMR1, ULPTMR2, ULPTMR3, ULPTIOC, ULPTISR, and ULPTCMSR.

Condition C: In count start mode or count restart mode (ULPTMR3.TEECTL[1:0] bits = 10 or 11)

When the counter is stopped (with the TCSTF bit of the ULPTCR register set to 0), even if 1 is set to the TSTART bit of the ULPTCR register and then the counter is started on an edge trigger of the ULPTEEn pin, the TCSTF bit remains at 0 (counter stopped) for five cycles of the count source.

Do not access the registers^{*1} related to the ULPT other than the TCSTF bit until the TCSTF bit is set to 1 (counter running).

When the counter is running (with the TCSTF bit set to 1), even if 0 (stop the counter) is set to the TSTART bit, the TCSTF bit remains at 1 (counter running) for five cycles of the count source.

Do not access the registers^{*1} related to the ULPT other than the TCSTF bit until the TCSTF bit is set to 0 (counter stopped).

Before changing the TSTART bit from 0 to 1, clear the interrupt register. For details, see [section 14, Interrupt Controller Unit \(ICU\)](#).

Note 1. Registers related to the ULPT: ULPTCNT, ULPTCMA, ULPTCMB, ULPTCR, ULPTMR1, ULPTMR2, ULPTMR3, ULPTIOC, ULPTISR, and ULPTCMSR.

25.5.2 Access to Flags (TUNDF, TCMAF, and TCMBF bits in ULPTCR Register)

Set 1 to the TUNDF, TCMAF, and TCMBF bits when changing the TSTART or TSTOP bits to prevent clearing TUNDF, TCMAF, or TCMBF by mistake. At this time, setting 1 to TUNDF, TCMAF, and TCMBF bits are ignored.

25.5.3 Access to the ULPTCNT, ULPTCMA, and ULPTCMB Registers

When setting to the same register sequentially while the counter is running (ULPTCR.TCSTF bit = 1), leave three or more cycles of count source between writes. The following registers are applicable:

- ULPTCNT
- ULPTCMA
- ULPTCMB

25.5.4 Mode Changes

Registers related to the operating mode of the ULPT (ULPTMR1, ULPTMR2, ULPTMR3, ULPTIOC, ULPTISR, and ULPTCMSR) can be changed only while the counter is stopped (both the TSTART and TCSTF bits of the ULPTCR register are set to 0). Do not change these registers while the counter is running.

If a register related to the operating mode of the ULPT is changed, the values of the TUNDF, TCMAF, and TCMBF bits of the ULPTCR register are undefined. Before starting the counter, write 0 to the TUNDF, TCMAF, and TCMBF bits.

- TUNDF (no underflow)
- TCMAF (no match for compare match A)
- TCMBF (no match for compare match B)

25.5.5 Setting the ULPTOn, ULPTOAn, and ULPTOBn Pins

To use the ULPTOn, ULPTOAn, and ULPTOBn pins as output pins, set up the ULPT, and then set the output in the PORT register after determining the initial output values.

25.5.6 Calculating the Number of Events

In event counter mode, the number of events is calculated using the following formula:

Number of events = Initial counter value (ULPTCNT register write value) – Counter value at the end of a valid event

25.5.7 When the Counter is Stopped Forcibly by the TSTOP Bit

Do not access the ULPTCNT, ULPTCMA, ULPTCMB, ULPTCR, ULPTMR1, ULPTMR2, or ULPTMR3 register during one cycle of the count source after stopping the counter forcibly by the ULPTCR.TSTOP bit.

25.5.8 Digital Filter

When using a digital filter, do not start the timer within five cycles of the digital filter clock after setting the ULPTIOC.TIPF[1:0] bits.

Also, when ULPTMR3.TEVPOL bit is changed while a digital filter is in use, do not start the timer within five cycles of the digital filter clock after setting the bit.

In Software Standby mode or Deep Software Standby mode 1, the digital filter clock (PCLKB) stops and the digital filter does not function. Before entering Software standby mode or Deep Software Standby mode 1, set the ULPTIOC.TIPF[1:0] bits to 00 (no filter).

25.5.9 Module-Stop Function Setting

ULPT operation can be disabled or enabled using Module Stop Control Register E (MSTPCRE). The ULPT module is initially stopped after reset. Releasing the module-stop state enables access to the registers.

25.5.10 Setting the ULPTEEn and ULPTEVIn Pins

To use the ULPTEEn and ULPTEVIn pins as input pins, set up the ULPT, and then set PmnPFS.PMR bit to 1.